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TFCG: A time-frequency enhanced deep learning model for temperature prediction in air-liquid cooled data centers

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HIGHLIGHTS

- We propose a novel TFCG model to achieve high-precision temperature predictions in data centers.
- The model extracts features from frequency and time domains via STFT and CNN, combined with GRU to capture short-term fluctuations and long-term trends.
- We develop a multi-channel approach incorporating correlation and thermodynamic analyses to minimize noise and enhance predictive accuracy.
- Extensive experiments on real-world datasets demonstrate that TFCG significantly outperforms state-of-the-art baseline models.

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ABSTRACT

Data center temperature prediction is essential for real-time cooling control and energy efficiency optimization. However, the multi-scale nature of temperature-related features in data centers poses significant challenges for accurate prediction. Existing approaches primarily rely on time-domain deep learning and often neglect frequency-domain features, while also adopting fully coupled inputs for multi-channel temperature signals, which limits their ability to capture complex cooling dynamics. To address these challenges, we propose a time-frequency enhanced deep learning model (TFCG) for temperature prediction in air-liquid cooling data centers. TFCG integrates time-frequency feature enhancement and long-range temporal modeling to capture multi-scale dynamic characteristics effectively. Moreover, channel-independent modeling with embedding-based fusion is introduced to handle complex thermal interactions. Comprehensive experiments on the Marconi100 dataset demonstrate that TFCG consistently outperforms strong state-of-the-art baselines, reducing MAE, RMSE, and MAPE by approximately 33.72%, 30.91%, and 10.75%, respectively, while improving R^2 by 3.05%. These results indicate significant improvements in both prediction accuracy and stability. These results provide important guidance for cooling optimization in hybrid data center environments.

1. Introduction

With the rapid advancement of big data and artificial intelligence technologies, global demand for computing power has increased dramatically. As the core supporting infrastructure, data centers have expanded significantly in scale. According to recent research, global data centers account for approximately 1.7% of total electricity consumption [1]. This trend not only exerts substantial environmental pressures but also incurs significant economic costs. Particularly, data center cooling systems consume approximately 40% of total energy usage [2]. Cooling systems account for a major portion of the overall energy consumption in data centers, and their optimized control is critical for improving overall energy efficiency. Temperature serves as a vital metric for evaluating both operational stability and energy efficiency in data centers. It directly impacts the reliability and lifespan of IT equipment (when temperatures are too high) and determines whether cooling energy consumption is efficient (when temperatures are too low). Therefore, establishing precise temperature prediction methods is a prerequisite for achieving optimized cooling system control and reducing overall energy consumption. However, the complex internal environment of

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data centers, including diverse spatial layouts, airflow organization, and the coupled effects of cooling control strategies, presents substantial challenges for achieving accurate and scalable temperature prediction.

In the past, substantial research efforts have been dedicated to modeling data center temperature prediction, yielding three distinct approaches: simplified/consolidated capacitance models, computational fluid dynamics/heat transfer (CFD/HT)-based models, and data-driven models [3]. The first two categories exhibit significant limitations in either accuracy or computational cost, making them ill-suited for real-time prediction requirements. In contrast, data-driven models strike a balance between efficiency and accuracy, gradually emerging as a research hotspot. In recent years, various architectures including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) have been extensively applied to temperature prediction tasks [4,5]. To better capture the long-range temporal dependencies and bidirectional contextual information in temperature sequences, more advanced base learners such as Long Short-Term Memory (LSTM) [6], Gated Recurrent Units (GRU) [7], and their bidirectional variants (BiLSTM [8] and BiGRU [9]) have been widely adopted. Despite the success of these temporal models and composite structures like CNN-BiLSTM [10] in representing spatiotemporal features, existing methods still exhibit shortcomings in channel modeling and multi-timescale feature extraction.

On one hand, mainstream approaches typically employ joint channel modeling, directly blending all data inputs into the model without adequately accounting for the physical independence and differences between channels [6]. This processing approach tends to induce mutual interference among high-dimensional features, which becomes more severe in the presence of noise or outliers [11], leading to a reduced sensitivity of the model to critical local patterns. On the other hand, data center temperature sequences often simultaneously exhibit short-term fluctuations (e.g., instantaneous load changes) and long-term trends (e.g., periodic load growth) [7]. Consequently, feature extraction must address components across different frequency scales [8]. Traditional deep learning models, however, struggle to capture such complex features [6]: they may be sensitive to short-term spikes but fail to preserve long-term dependencies, or they can depict overall trends while losing local details. This results in prediction biases and insufficient generalization capabilities.

To address these challenges, this paper proposes a multi-channel deep prediction model called TFCG, which combines time-frequency enhancement with long-range temporal dependency modeling. This model jointly extracts multi-scale time-frequency features through Short-Time Fourier Transform (STFT) and Convolutional Neural Networks (CNN), utilizes Gated Recurrent Units (GRU) to capture long-range temporal dependencies, and incorporates both channel-independent and channel-embedding mechanisms to effectively reduce cross-channel interference. In summary, the core contributions of this paper are as follows:

(1) We propose a TFCG model that integrates time-frequency enhancement with long-range temporal dependency modeling. By jointly processing the frequency and time domains, it effectively captures both short-term fluctuations and long-term trend characteristics in temperature sequences.
(2) We explore the relationships between temperature features in data centers, incorporating correlation analysis and thermodynamic factors. Through a multi-channel approach, where each channel predicts the input temperature of different racks using relevant correlated features, we minimize noise interference and enhance predictive accuracy.
(3) Extensive experiments on real-world data center datasets demonstrate that the proposed method consistently outperforms existing approaches in terms of both prediction accuracy and stability for data center temperature forecasting.

The remainder of this paper is organized as follows. Section 2 reviews the research progress and challenges in related work. Section 3 outlines the research methodology. Section 4 presents the experimental results and analysis. Section 5 concludes the study.

2. Related work

In recent years, data center temperature prediction has attracted extensive attention from both academia and industry, making significant contributions to improving thermal safety and energy efficiency. In this section, we provide a structured review of existing methods, including physics-based thermal models that rely on thermodynamic principles to describe heat transfer processes, data-driven thermal modeling approaches, and related research in time-series forecasting methods.

2.1. Physics-based thermal modeling methods

Early research on data center thermal modeling primarily established mathematical expressions for target objects based on physical principles (such as thermodynamic laws and the law of mass conservation). Tang et al. [12] employed the POD-Galerkin projection method to obtain POD temperature coefficients and calculated POD velocity coefficients via POD interpolation to predict three-dimensional temperature fields. Comparisons with CFD simulations demonstrated that the POD method effectively predicts fluid flow and temperature fields across scales. Guo et al. [13] established a 3D model of a data center, defined its mesh discretization and boundary conditions, and employed CFD simulations to analyze airflow within the facility.

Subsequently, these approaches evolved into hybrid models, which integrate physical laws with data obtained from experiments or simulations to approximate model parameters. Hashemi [14] et al. pioneered the application of physics-informed neural networks (PINNs) to model data center environments. This approach demonstrates significant accuracy improvements. Parameterized PINNs are developed to handle varying operating conditions, enhance prediction efficiency and significantly reduce computational costs. Zhang et al. [15] introduced Feature Integration Machine Learning (FIML), combining CFD features with sparse measured data to achieve rapid prediction of critical point temperatures. Despite their strong physical interpretability and improved reliability, these models still face several limitations when applied to large-scale and highly dynamic data center environments. Their performance often relies on the availability of accurate physical priors or CFD simulations, which can be computationally expensive and difficult to generalize across different layouts and operating conditions.

2.2. Data-driven based thermal modeling methods

Concurrently, the field has seen an increasing trend in leveraging data-driven frameworks. Researchers utilize appropriate mathematical models such as regression statistics or neural networks to fit nonlinear input–output mapping relationships without relying on explicit physical mechanism modeling. Based on system data analysis, they employ training data to build thermal models capable of performing real-time temperature prediction. Ilager et al. [16] employed an XGBoost model combined with sensor data to train a temperature predictor, designing a dynamic scheduling algorithm based on it. Wang et al. [8] proposed a CNN-BiLSTM-Attention neural network model to achieve multiscale simulation of data centers. Mao et al. [17] proposed a novel hybrid neural network model (BiTCN-Attention) and designed a dynamic model updating framework for dynamic thermal management in data center buildings. Chen et al. [18] introduced an Adaptive Physically Consistent Neural Network (A-PCNN) framework. The model employs NN and Softplus activation functions, replacing traditional predetermined and fixed coefficients to reduce trial-and-error costs and enhance flexibility. Such models are typically sensitive to high-dimensional, multi-channel inputs, susceptible to interference between channels, and struggle to fully capture multi-timescale features encompassing both short-term fluctuations and long-term trends.

2.3. Time sequence modeling methods

While mainstream deep learning models have achieved notable success, they often struggle with long-term volatility and the stochastic noise inherent in large-scale data center environments. To address these challenges, researchers have begun exploring advanced frontier architectures. Zhou et al. [19] proposed a Transformer-based thermal surrogate model, where self-attention is used for capturing the temporal and spatial characteristics in the temperature field to replace CFD. In the general field of time-series forecasting, research has revealed [20] that Transformer-based approaches are not only computationally expensive but also struggle to capture a global view of the time series. To address this issue, Zhou et al. [20] proposed a Frequency Enhanced Decomposed Transformer (FEDformer), which is more efficient than the standard Transformer with a linear complexity relative to the sequence length. More recently, the paradigm has shifted toward patch-based modeling and channel-independent architectures to address the limitations of traditional point-wise time-series analysis. Nie et al. [21] proposed an efficient design of Transformer-based models for multivariate time series forecasting and self-supervised representation learning. The channel-independent patch time series Transformer (PatchTST) can improve the long-term forecasting accuracy significantly when compared with that of SOTA Transformer-based models. However, while the effectiveness of frequency-domain analysis and channel-independence has been demonstrated in the general field of time-series forecasting, these techniques remain underexplored in the context of data center thermal management.

Most deep learning models for time-series forecasting focus on capturing temporal dependencies, and several recent approaches have incorporated frequency-domain representations to model long-term dependencies and periodic patterns. For example, models such as FEDformer [20] leverage frequency-domain mechanisms to enhance multi-scale feature extraction. However, these methods typically adopt joint modeling across all variables, directly mixing multi-channel inputs without considering the underlying physical characteristics of the system. In data center environments, rack-level temperature dynamics exhibit partial independence, while being only weakly coupled through airflow and thermal interactions. Under such conditions, joint modeling may introduce cross-channel interference and degrade prediction performance, especially in high-dimensional settings. Meanwhile, some studies have attempted to alleviate issues arising from high-dimensional inputs, such as applying filters to remove noise interference [22] or using sparse autoencoders for dimensionality reduction [23]. In addition, recent works such as PatchTST [21] explore channel-independent modeling strategies. However, these approaches are primarily developed for general-purpose time-series forecasting, where channels are treated in a uniform manner without considering the task-specific characteristics of the system. In contrast, rack-level temperature prediction in data centers exhibits structured heterogeneity, where each rack is influenced by a distinct subset of thermal and operational factors.

To address these limitations, this paper proposes a multi-channel deep learning forecasting model (TFCG). Unlike existing methods, the proposed model integrates physically motivated channel-independent modeling with explicit time-frequency feature enhancement. Specifically, a channel-independent strategy is adopted to model each rack individually based on its most relevant features, thereby reducing inter-channel interference while preserving intrinsic thermal dynamics. In addition, an explicit time-frequency enhancement module based on STFT and CNN is introduced to capture both localized frequency patterns and temporal variations. Combined with GRU-based sequence modeling for long-term dependency learning, the proposed model is able to effectively represent both short-term fluctuations and long-term periodic behaviors. By jointly considering physical consistency, channel heterogeneity, and multi-scale temporal-frequency characteristics, the proposed approach provides a structured solution for modeling complex thermal dynamics in data centers, leading to more accurate and reliable rack temperature prediction.

3. Model design

3.1. Model framework

Accurate temperature prediction in data centers plays a crucial role in enabling real-time control and energy efficiency optimization of cooling systems. This study focuses on rack-level temperature prediction, which is a key indicator for maintaining server operational stability and optimizing cooling regulation. Fig. 1 presents the overall framework of the thermal prediction modeling process for data centers, where the data center is illustrated using the layout of the data source for visualization purposes. The input data consist of historical monitoring records from the data center, including rack inlet temperature, rack power consumption, and features from both air-conditioning and liquid-cooling systems. To improve prediction accuracy, the input data are first processed using Granger causality analysis [24] to identify and retain the most relevant features. Subsequently, the proposed model is trained and validated on a pre-segmented dataset, ultimately yielding the predicted rack temperature results.

To fully exploit both time-domain and frequency-domain information, this paper proposes a multi-channel deep prediction model based on TFCG. Fig. 2 provides a detailed view of the TFCG prediction model structure. The model adopts a channel-independent strategy and extracts multi-scale features from both the frequency and time

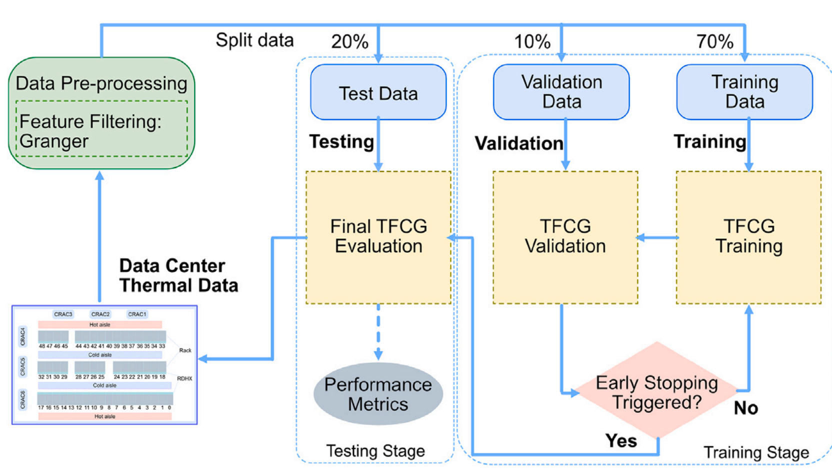


Fig. 1. Thermal prediction modeling framework for data centers.

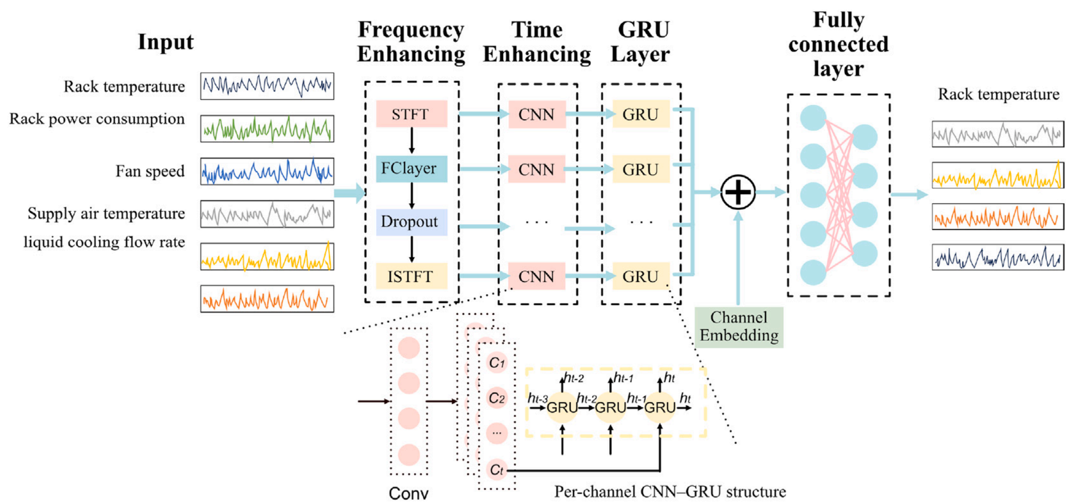


Fig. 2. Structure of the TFCG prediction model.

domains through time-frequency enhancement. It utilizes GRU to model long-term dependencies across time steps. Finally, a channel embedding module integrates features from all channels, and a linear layer generates the rack inlet temperature prediction. By integrating time-frequency feature extraction with channel-independent modeling, this model effectively captures multi-scale dynamic characteristics while reducing inter-channel interference, achieving high-precision predictions.

3.2. Data preprocessing

Physical Mechanisms. In data centers, temperature serves as the core parameter of cooling control, exerting a substantial impact on overall energy efficiency [25]. However, rack temperature is influenced by multiple interacting factors and exhibits complex spatiotemporal dynamics, making accurate modeling particularly challenging. From a physical perspective, rack temperature evolution is governed by a coupled heat transfer process involving heat generation, convective cooling, and fluid heat exchange [26]. Specifically, the heat generated by equipment, is dissipated through airflow and liquid cooling loops. The supply air temperature of Computer Room Air Conditioning (CRAC) units directly determines the cooling capacity, while fan speed affects the convective heat transfer efficiency. Meanwhile, the Rear Door Heat Exchangers (RDHx) remove heat via liquid cooling, where the coolant flow rate influences the heat exchange intensity. In addition, rack temperature is not only affected by its own thermal behavior but also influenced by spatial thermal interactions. Due to airflow recirculation and imperfect containment, heat from neighboring racks may propagate to the target rack, leading to thermal coupling effects. Therefore, rack temperature can be characterized as a nonlinear dynamic process jointly affected by internal heat generation, cooling system control variables, and inter-rack thermal interference.

Therefore, unlike many heat prediction methods that primarily rely on server-side operational metrics [27], we account for the fact that rack inlet temperature is influenced not only by server heat generation but also by complex cooling mechanisms such as air conditioning loops and liquid cooling circuits. The dataset used in this study is derived from the Marconi100 data center [28], which adopts a hybrid air-liquid cooling architecture integrating CRAC units and RDHx. It contains fine-grained historical monitoring records with multiple thermal and operational variables, including rack temperature, rack power consumption, fan speed, supply air conditions, and liquid cooling flow rate. These variables correspond directly to key physical processes such as heat generation, convective cooling, and fluid-based heat exchange. As a result, it preserves the underlying thermodynamic relationships of the system, enabling the proposed model to learn physically meaningful patterns for accurate temperature prediction.

Stochastic Disturbances. In data center operations, stochastic disturbances such as equipment faults do not typically lead to abrupt temperature spikes. Due to thermal inertia and the compensatory effects of redundant cooling systems (e.g., multiple air conditioners and liquid cooling loops), the impact of a single device failure is often reflected as a gradual temperature variation rather than an instantaneous change. Extreme scenarios (e.g., simultaneous failure of all cooling units) are rare in practice, as they correspond to critical system-level faults. Therefore, instead of explicitly modeling all possible stochastic disturbances, their effects are implicitly captured through the temporal modeling framework. Although such disturbances are inherently random, their influence on rack temperature is often delayed due to thermal inertia, resulting in smooth trend variations in the time-series data. The proposed model integrates convolutional neural networks (CNN) and short-time Fourier transform (STFT) to capture both temporal and frequency-domain patterns, while GRU is employed to model long-term temporal dependencies. By learning from historical data containing diverse operational variations, the model can capture both transient fluctuations and longer-term patterns in temperature dynamics. This design enables the model to achieve stable and accurate temperature prediction under diverse operational conditions.

Feature Selection. As illustrated in Fig. 3, which presents the monitoring data from the Marconi100 data center in May 2021, the temperature distributions across representative racks exhibit significant spatial heterogeneity in terms of variance, median values, and

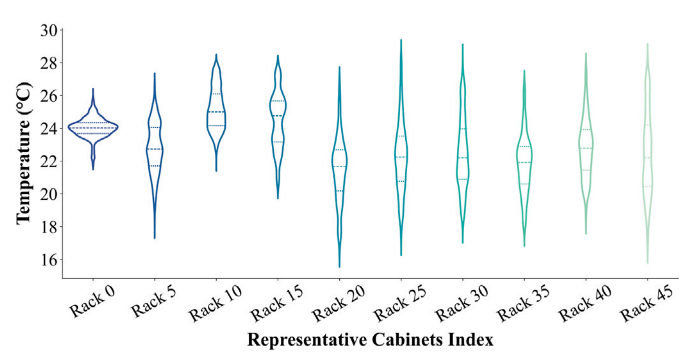


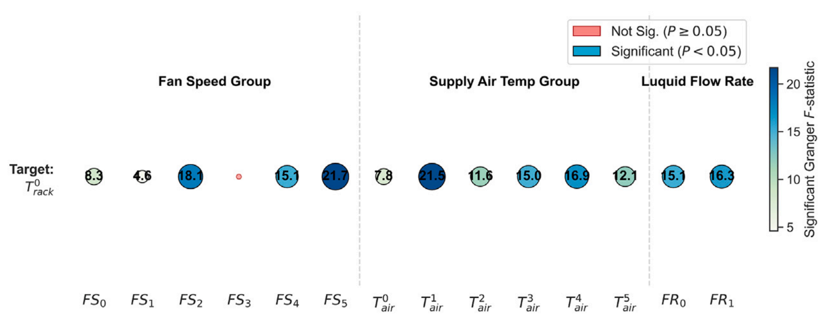
Fig. 3. Spatial heterogeneity analysis of rack cabinet temperature distributions.

modality. This diversity suggests that cooling efficiency and thermal dynamics vary substantially across physical locations, making a uniform prediction model insufficient for all racks. To address this, we implement a multi-channel architecture to model racks independently, which effectively captures these localized characteristics. However, incorporating too many variables for each channel can increase model complexity and redundancy. Consistent with previous studies [3], we employ temperature-related features for rack temperature prediction. Given that the original dataset contains a large number of variables, many of which may have limited relevance to rack temperature, we apply Granger causality analysis to perform feature grouping and selection. The Granger causality test first calculates the p-value for each feature group. If the p-value is less than 0.05, indicating a statistically significant causal relationship, the corresponding F-statistic is further considered to assess the strength of the relationship. Specifically, the features are categorized into fan speed, supply air temperature, liquid cooling flow rate, and adjacent rack temperature. Then, Granger causality analysis is performed within each category to determine the most relevant features for each rack. This process allows us to retain only features with a significant causal impact on rack temperature, thus reducing redundancy while ensuring model interpretability.

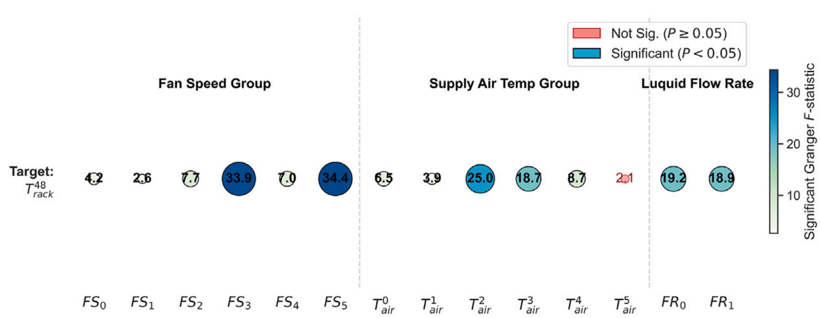
As shown in Fig. 4(a) and Fig. 4(b), Rack 0 and Rack 48 are taken as representative examples to illustrate the feature selection process via bubble-map visualization. For Rack 0, the p-value of FS_3 (representing the fan speed of the 3rd air conditioner) is greater than 0.05, indicating a weak causal relationship with the rack temperature. Among the remaining significant variables, FS_5 exhibits the largest F-statistic, suggesting the strongest causal influence, and is therefore selected. Similarly, for the supply air temperature and liquid cooling flow rate groups, T^1_{air} and FR_1 are selected, respectively. Following the same selection criterion for Rack 48, the fan speed FS_5, supply air temperature T^2_{air}, and liquid cooling flow rate FR_0 are identified as the most relevant features.

Inspired by previous research [29], we also consider rack power consumption, which has a direct effect on rack temperature. Therefore, power features from each rack are incorporated into the final predictive model. To further validate the necessity of incorporating rack power consumption and neighboring rack temperature into our feature set, we employed the XGBoost model coupled with SHAP analysis [29] to quantify their contributions from a model-interpretability perspective. As illustrated in Fig. 5, rack power consumption (P_{rack}) consistently emerges as the most dominant predictor for both Rack 0 and Rack 48. This confirms that P_{rack} is the primary proxy for internal heat generation, and its inclusion is fundamental for capturing the core thermal load.

Furthermore, the neighboring rack temperature ($T_{neighbor}$) also ranks among the top three most influential features in both cases. For Rack 0 (Fig. 5a), $T_{neighbor}$ is the second most important factor, while for Rack 48 (Fig. 5b), it remains highly significant despite the stronger influence of supply air temperature (T^5_{air}). It should be noted that while T^5_{air} exhibits high importance in the SHAP analysis for Rack 48, it was not included in the final feature set because it failed to meet the linear causality threshold ($p < 0.05$) in the Granger test. This discrepancy suggests that while T^5_{air} may contain non-linear predictive information, the Granger-based selection prioritizes variables with robust, direct causal links to maintain model parsimony. The high importance of $T_{neighbor}$ across different rack locations validates the existence of strong inter-rack thermal coupling. It demonstrates that a rack's thermal state is not isolated but is heavily influenced by the heat dissipation of its neighbors. These results justify the inclusion of both power consumption and spatial coupling features, as they represent the essential internal and external heat-transfer mechanisms within the data center. Based on the above feature selection and interpretability analysis, each sample in this study is constructed as a multivariate input–output sequence pair using a sliding window strategy. The input consists of a historical sequence of selected features

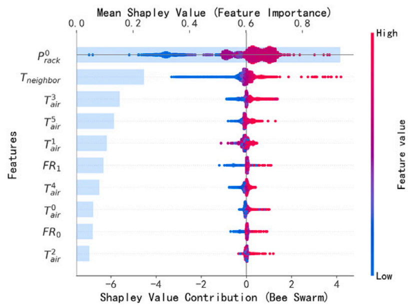


(a) Selection results for Rack 0.

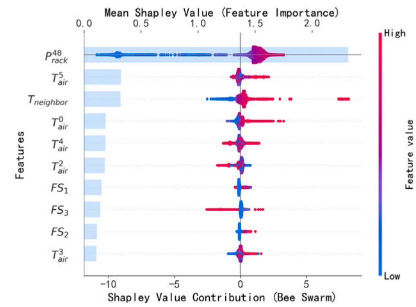


(b) Selection results for Rack 48.

Fig. 4. Feature importance based on Granger causality analysis. The bubbles visualize the causal strength between various input parameters and the target rack temperature. The bubble size and color intensity represent the magnitude of the F-statistic, while red markers indicate features with no statistically significant causal relationship ($P \geq 0.05$). FS_i and T^i_{air} represent the fan speed and supply air temperature of the i-th air conditioner, respectively; FR_i indicates the flow rate of the i-th liquid cooling loop.

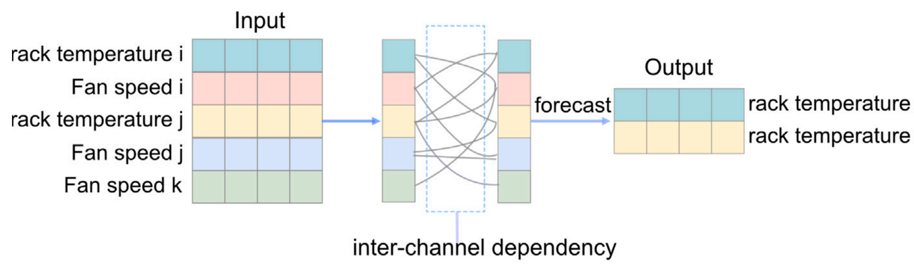


(a) Importance analysis of SHAP features of Rack 0

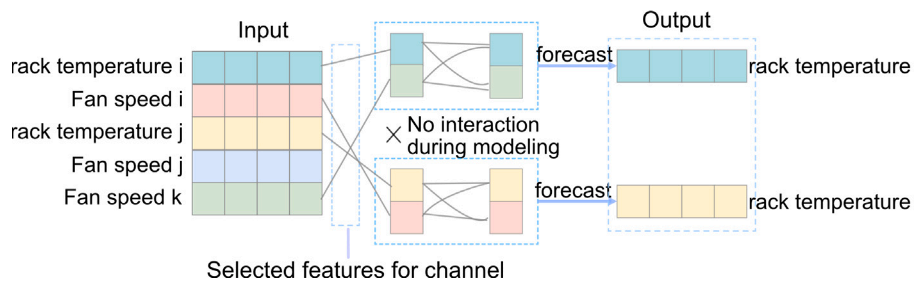


(b) Importance analysis of SHAP features of Rack 48

Fig. 5. Feature importance and interpretability analysis using SHAP.



(a) Channel Dependent Strategy



(b) Channel Independent Strategy

Fig. 6. Illustration of two different prediction strategies, taking fan speed as an example of influencing factors. (a) Channel-dependent strategy: All rack temperatures and fan speeds are jointly modeled within a unified channel, where fan speeds from different racks may introduce inter-channel interference and affect the prediction of a target rack temperature. (b) Channel-independent strategy: For each target rack temperature, only the most relevant fan speed features are selected for modeling, while irrelevant fan speeds are excluded, thereby reducing inter-channel interference.

for each rack, including rack temperature, rack power consumption, fan speed, supply air temperature, liquid cooling flow rate, and neighboring rack temperature. The output corresponds to the future rack temperature over a multi-step prediction horizon.

Channel Independent Strategy. Existing data center thermal prediction models are typically channel-dependent, where interactions among multiple input variables are jointly modeled (as shown in Fig. 6(a). However, such approaches may introduce inter-channel interference and degrade prediction performance. Recent studies [21,30] have shown that channel-independent modeling can effectively improve forecasting accuracy. Motivated by these findings, we adopt a channel-independent strategy (as shown in Fig. 6(b), where each channel (i.e., each rack) is modeled independently using its most relevant features. This design allows the model to better capture intra-channel temporal dependencies while reducing interference from irrelevant variables. For clarity, Fig. 6(b) illustrates the concept using fan speed as an example, while in practice multiple heterogeneous variables (e.g., supply air temperature and other related parameters) are considered. To enable independent processing, a channel-wise partitioning is performed during data preprocessing, ensuring that each rack's feature set is separately organized and fed into the subsequent prediction model.

Instance Normalization. Data center workloads are typically time-varying, which causes significant fluctuations in temperature data [31]. To mitigate these effects, we adopt a simple normalization strategy inspired by instance normalization. Specifically, for each input sequence, we subtract the last observed value from all time steps, allowing the model to focus on relative temporal variations rather than absolute levels. The transformation is defined as:

$$\mathbf{x}_{1:L}^{(i)} = \mathbf{x}_{1:L}^{(i)} - x_L^{(i)}, \tag{1}$$

and the predicted outputs are restored to the original scale by:

$$\hat{\mathbf{y}}_{L+1:L+H}^{(i)} = \hat{\mathbf{y}}_{L+1:L+H}^{(i)} + x_L^{(i)}, \tag{2}$$

where $x_L^{(i)}$ denotes the last observed value of the i-th channel. This operation effectively reduces distribution shifts and improves the model's ability to capture relative temporal dynamics.

3.3. Prediction network architecture

After data preprocessing, constructing an effective prediction network is essential for extracting multi-scale features to achieve high-precision rack temperature forecasting in data centers. Rack temperature prediction can be formulated as a multivariate nonlinear time-series forecasting problem. Due to thermal inertia and heat transfer delay effects, spatial correlations induced by inter-rack thermal coupling, as well as the highly nonlinear interactions among heterogeneous factors such as airflow conditions and liquid cooling parameters, it is difficult to derive accurate analytical models solely based on physical equations. Deep learning models provide a data-driven alternative for such problems. On one hand, recurrent neural networks (e.g., GRU) are capable of capturing long-term temporal dependencies. On the other hand, time-frequency analysis methods (e.g., STFT combined with CNN) can effectively extract multi-scale temporal patterns, including periodic fluctuations and transient variations. Based on these observations, we design a channel-independent prediction model (TFCG), where each rack is modeled separately to reduce cross-channel interference while preserving temporal dynamics. Finally, the fused representation is passed through a linear output layer to generate the predicted rack temperatures. The overall architecture of the proposed model is illustrated in Fig. 2, and the following sections provide a detailed breakdown of its key components.

3.3.1. Time-frequency enhancement block

To effectively capture both periodic and short-term variations in data center thermal dynamics, we design a time-frequency enhancement block that integrates frequency-domain decomposition with temporal feature refinement.

Frequency enhancement. Given an input multivariate sequence X, we first apply the Short-Time Fourier Transform (STFT) to obtain a joint time-frequency representation:

$$X_s = \mathrm{STFT}(X), \tag{3}$$

where X_s encodes localized frequency components over time, enabling the modeling of non-stationary thermal behaviors.

To provide a clearer understanding of the spectral features captured by our model, Fig. 7 visualizes the transformation of a raw temperature sequence into a frequency map. The input sequence (Fig. 7, left) displays the non-stationary thermal fluctuations of a specific rack over 1000 time steps. Through the STFT module with a window length of $N_{fft} = 8$, the signal is decomposed into multiple frequency bins (Fig. 7, right). The low-frequency components (Bin 0-Bin 1) exhibit higher magnitude values, indicating dominant slow-varying thermal dynamics and steady-state cooling behavior. In contrast, higher-frequency components (Bin 2-Bin 4) capture short-term transient variations, such as abrupt thermal fluctuations. This representation provides a frequency-aware characterization of temperature dynamics.

To enhance discriminative frequency representations, a lightweight feed-forward network is applied:

$$Z_t = \mathrm{FFN}(X_s). \tag{4}$$

This operation adaptively reweights frequency components to highlight dominant thermal fluctuation patterns. The enhanced representation is then mapped back to the time domain using the inverse STFT:

$$Z = \mathrm{ISTFT}(Z_t), \tag{5}$$

resulting in a frequency-domain feature representation enriched with periodic and multi-scale frequency information.

To further analyze the impact of STFT parameter configurations, we evaluated different window functions, including Hann, Hamming, and rectangular windows, together with different window sizes (4, 8, and 16). The experimental results demonstrate that the rectangular window with a window size of 4 provides the most effective balance between temporal localization and frequency representation for non-stationary thermal signals. Compared with Hann and Hamming windows, the rectangular window preserves abrupt thermal variations more effectively, which is beneficial for capturing short-term temperature dynamics in data center environments. Accordingly, the rectangular window with a window size of 4 was selected as the final configuration of the proposed model. Detailed experimental comparisons are presented in Appendix (see Table A.5).

Time enhancement. After time-frequency enhancement, features from each channel are further refined using a lightweight 1D convolutional module for time-domain enhancement [32].

The input is denoted as $X_c^{in} \in R^{f \times L}$, where f refers to the number of features per channel. The computational process of the channel-independent convolution module is described as follows:

$$X_c^{cnn} = \mathrm{BN}(\mathrm{ReLU}(\mathrm{Conv1D}(X_c^{in}))), \tag{6}$$

$\mathrm{Conv1D}(\cdot)$ denotes a one-dimensional convolution operation; $\mathrm{ReLU}(\cdot)$ represents the rectified linear unit activation function; $\mathrm{BN}(\cdot)$ denotes batch normalization. The output length of the convolution is determined by the kernel size, stride, and padding. This architecture ensures that each channel effectively learns its optimal local feature representation without interference from other channels.

3.3.2. Gated recurrent unit

To capture long-range temporal dependencies in thermal evolution, we employ a Gated Recurrent Unit (GRU) for sequence modeling.

For each channel c, the encoded sequence is modeled as:

$$H_c = \mathrm{GRU}(X_c^{cnn}) \in R^{L' \times h}. \tag{7}$$

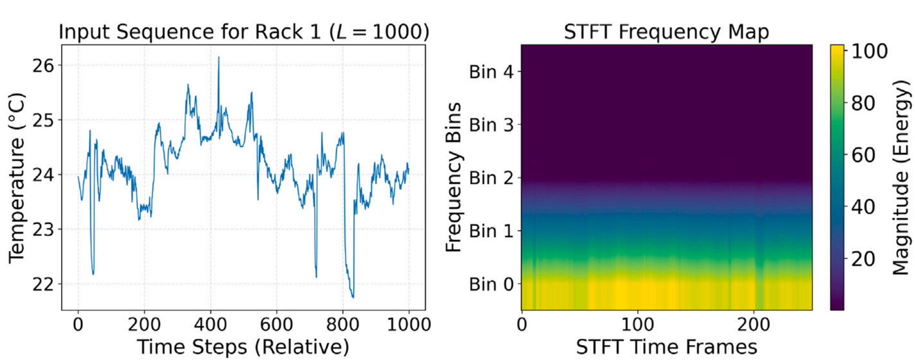


Fig. 7. Transformation from raw temperature sequence to frequency-domain representation. Left: A 1000-step raw thermal signal showing non-stationary fluctuations. Right: The STFT-based frequency map ($N_{fft} = 8$).

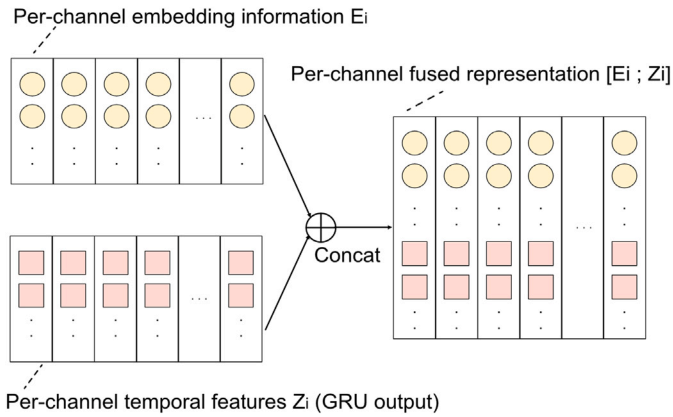


Fig. 8. Structure of the channel embedding module.

To obtain a compact representation, temporal average pooling is applied:

$$Z_c = \frac{1}{L'} \sum_{t=1}^{L'} H_C^{(t)}. \tag{8}$$

Finally, the representations of all channels are concatenated and fed into the subsequent module.

3.3.3. *Channel embedding*

Under the channel-independent strategy, each channel acquires an independent temporal representation via dedicated CNN and GRU modules. To enable the model to effectively exploit multi-channel information, it is necessary to inject channel-specific information that distinguishes the physical characteristics of each channel. To this end, we design a channel embedding mechanism, which assigns a unique embedding vector to each channel, thereby bridging independent modeling and unified fusion. Specifically, for a multichannel input time series, each channel i is associated with a learnable embedding vector E_i obtained from a shared embedding lookup table indexed by the channel identifier. Based on this design, the channel embedding process can be formulated as follows:

$$\begin{aligned} E_i &= \text{Embedding}(i), \\ E &= [E_1; E_2; \dots; E_C], \end{aligned} \tag{9}$$

where [;] denotes concatenation along the channel dimension. The structure of the channel embedding module and its integration process are illustrated in Fig. 8. As shown, the per-channel embedding information E_i is concatenated with the corresponding GRU output sequence representation Z_C (denoted as Z_i for each channel), thereby integrating channel-specific information with the independently learned temporal features:

$$F = \text{Linear}(\text{Concat}(E, Z_C)), \tag{10}$$

The fused representation F retains the local temporal dynamics captured by the CNN-GRU and encodes each channel's physical attributes via the channel embeddings, thus providing the unified model with a principled basis for distinguishing among channels.

After obtaining the fused features, the fused features are passed through the output layer (a fully connected output layer) to generate the final multi-step prediction:

$$\begin{aligned} \hat{Y} &= \text{OutputLayer}(F), \\ Y &= InstanceNormalization^{-1}(\hat{Y}), \end{aligned} \tag{11}$$

where Y represents the final prediction. The overall procedure of TFCG is outlined in Algorithm 1.

Algorithm 1: TFCG Algorithm.

Input: Historical data center data $\mathbf{X} \in \mathbb{R}^{L \times F}$
Output: Future prediction $\mathbf{Y} \in \mathbb{R}^{H \times C}$

```
Initialization: Look-back window L, horizon H, feature size F, channel C;
X ← Instance Normalization(X);
X_s ← STFT(X);
Z_t ← FFN(X_s);
Z ← ISTFT(Z_t);
for c = 1 to C do
    X_c^cnn ← BN(ReLU(Conv1D(X_c)));
    H_c ← GRU(X_c^cnn);
    Z_c ← GlobalAveragePooling(H_c);
end
E ← LearnableChannelEmbedding(C);
F ← Linear(Concat(E, Z));
Ŷ ← OutputLayer(F);
Y ← Instance Normalization^{-1}(Ŷ);
return Y;
```

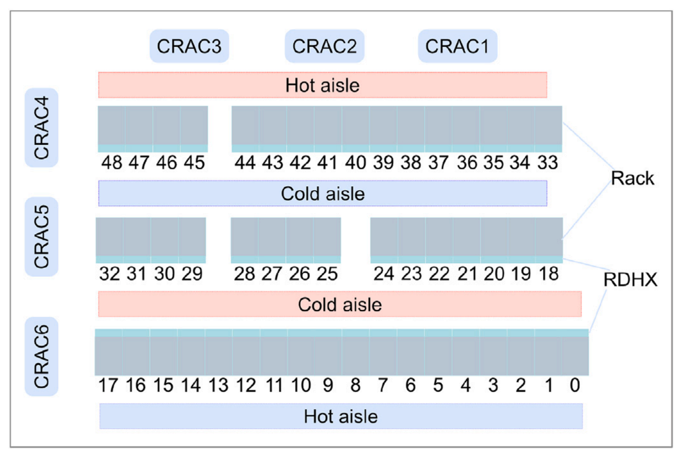


Fig. 9. Layout of the Marconi 100 data Center.

4. Experiments and results

4.1. *Experimental setup*

4.1.1. *Dataset*

The experiments were conducted using the publicly available Marconi100 dataset provided by CINECA [28]. This dataset encompasses comprehensive system-level monitoring information, ranging from internal computing node metrics such as core load, temperature, frequency, CPU power consumption, and fan speed to infrastructure-level parameters, including liquid cooling and air conditioning system data. The cooling configuration of the data center employs a hybrid cooling scheme, combining rear-door liquid cooling with traditional air cooling. Specifically, six precision air conditioners and two liquid-cooling circuits are deployed in the server room to supply cooling for servers equipped with GPUs. The rack layout of the data center is illustrated in Fig. 9. The dataset covers the period from January 4, 2020, to January 9, 2022, and contains approximately 40,000 raw time-series data points collected at 15-minute intervals. The data were divided into training, validation, and test sets in a ratio of 7:1:2.

4.1.2. *Evaluation metrics*

To evaluate model performance, five commonly used evaluation metrics were adopted: Mean Absolute Error (MAE), Root Mean Squared

Error (RMSE), Mean Absolute Percentage Error (MAPE), the coefficient of determination (R^2) and an efficiency metric, average inference time per sample during testing (ms/sample). The corresponding calculation formulas are as follows:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \hat{y}_i\right|, \tag{12}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \tag{13}$$

$$\mathrm{MAPE} = \frac{100\%}{N}\sum_{i=1}^{N}\left|\frac{\hat{y}_i - y_i}{y_i}\right|, \tag{14}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}, \tag{15}$$

$$\text{Inference Time} = \frac{\text{Total testing time (ms)}}{\text{Number of samples}}, \tag{16}$$

where N denotes the number of input–output pairs provided to the model during testing, and $\hat{y}_i$ and y_i represent the predicted value and the observed (true) value, respectively.

4.1.3. Baseline model

To comprehensively evaluate the effectiveness of the proposed TFCG model, we select twelve representative baseline methods covering classical statistical models, machine learning methods, and recent deep learning-based time-series forecasting approaches.

- ARIMA: A classical statistical forecasting model that captures linear temporal dependencies.
- ANN: A multilayer perceptron model applied in an autoregressive manner for multi-step prediction.
- XGBoost: A gradient boosting decision tree model for nonlinear regression tasks.
- LSTM: A recurrent neural network designed to model long-term temporal dependencies.
- BiLSTM: A bidirectional extension of LSTM that captures both forward and backward temporal contexts.
- BiGRU: A bidirectional gated recurrent unit model with improved computational efficiency.
- Attention-DNN [33]: A data-driven server temperature prediction model combining numerical simulations with deep learning. It uses a deep neural network to train on the simulation-generated temperature data, and incorporates an attention mechanism to capture important feature information during training.
- Transformer [19]: A thermal surrogate model based on the Transformer architecture, which captures spatiotemporal features of the temperature field via self-attention as a substitute for CFD simulations, and proposes an SAC-based framework using the surrogate model.
- CNN-BiLSTM-Attention [8]: A method combining multi-scale collaborative modeling and deep learning. It constructs multi-scale data center models by using the parent model's simulation results as boundary conditions for the sub-model and proposes a CNN-BiLSTM-Attention network to predict maximum CPU temperatures.
- BiTCN-Attention [17]: A hybrid model integrating bidirectional temporal convolution and attention mechanisms for multiscale temporal feature extraction.
- FEDformer [20]: A frequency-enhanced Transformer that incorporates decomposition and frequency-domain learning for long-term forecasting.
- PatchTST [21]: A patch-based Transformer with channel-independent design for efficient long-sequence modeling.

The performance of the proposed TFCG and the baseline models was comprehensively evaluated in real data center conditions. All models were implemented using PyTorch 1.13.1. Furthermore, to maintain conciseness, the detailed parameter configurations and intermediate tensor shapes of the proposed TFCG model and all baseline models during forward propagation are provided in Appendix (see Table A.6). The reported configurations include key hyperparameters, layer structures, and feature dimensions of intermediate representations to improve the reproducibility of the experiments and clarify the tensor transformations throughout the networks.

4.2. Experimental results and analysis

4.2.1. Comparison of model prediction performance

We compared the predictive accuracy of our proposed TFCG model with that of other methods using real-world data. Prior thermal modeling work [6] has demonstrated that data center temperature exhibits significant dynamic behavior that must be captured in prediction models, rather than being treated as purely steady state. Accordingly, a 6-step prediction horizon (corresponding to 90min at a 15-minute sampling interval) is adopted, as it is sufficiently long to capture the dominant transient thermal response while avoiding excessive uncertainty associated with longer horizons.

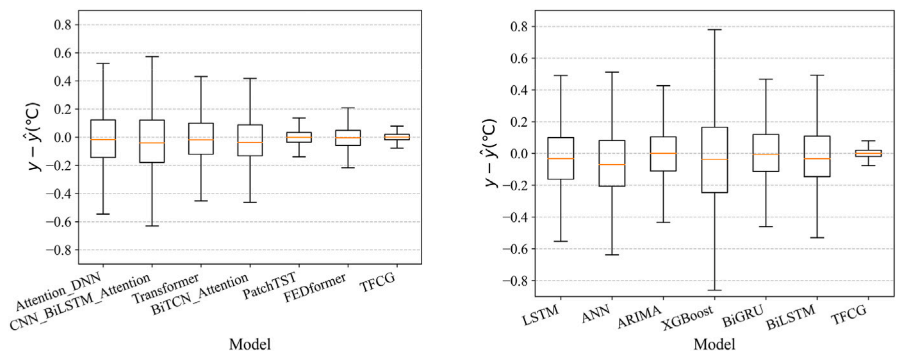
For Attention-DNN and CNN-BiLSTM-Attention, which explicitly employ Bayesian Optimization (BO) for hyperparameter tuning in their original studies, special care was taken to ensure a fair comparison under the 6-step forecasting horizon. Specifically, we evaluated these two models under multiple configurations, including the best hyperparameters reported in their original papers, the configurations consistent with the proposed TFCG model, and additional hyperparameter sets obtained through more than ten BO-based search trials. The best-performing results among all configurations were selected as the final comparison results. For all other baseline models, hyperparameters were kept consistent with those of the proposed model to ensure a fair comparison. All evaluation results are summarized in Table 1. All experiments were repeated five times with different random seeds, and the results are reported as mean ± standard deviation. It is worth noting that ARIMA and XGBoost exhibit zero variance across runs. For ARIMA, this is due to its deterministic estimation process given fixed data and model parameters. For XGBoost, although randomness can be introduced through subsampling, these stochastic components were fixed in our implementation, resulting in deterministic behavior and identical predictions across runs.

Performance Comparison. The results in Table 1 demonstrate that the proposed TFCG model consistently achieves the best performance across all evaluation metrics. Compared with both traditional methods and recent time-series forecasting models, TFCG shows significant improvements in prediction accuracy. Even when compared with advanced forecasting models such as PatchTST and FEDformer, TFCG still maintains clear advantages. Specifically, compared with PatchTST, TFCG reduces MAE, RMSE, and MAPE by 33.72%, 30.91%, and 10.75%, respectively, while improving R^2 by 3.05%. Similarly, compared with FEDformer, TFCG achieves reductions of 45.85% in MAE, 34.70% in RMSE, and 23.78% in MAPE, along with a 3.77% improvement in R^2. It is also worth noting that some complex hybrid models do not necessarily outperform simpler sequence models (e.g., BiGRU), suggesting that effective modeling of temporal dependencies and feature relevance is more critical than increasing architectural complexity. In terms of computational efficiency, although the inference time of TFCG (18.7868 ms/sample) is slightly higher than that of lightweight models such as ANN and LSTM, it remains comparable to PatchTST and significantly lower than FEDformer. This demonstrates that TFCG achieves a favorable trade-off between prediction accuracy and computational cost. Overall, these results validate the effectiveness of the proposed model in capturing complex thermal dynamics, benefiting from the integration of time-frequency feature extraction, channel-independent modeling, and long-term temporal dependency learning.

The superior performance of TFCG mainly benefits from its physically motivated design for modeling complex thermal dynamics in

Table 1
Prediction performance comparison of different models under a 6-step forecasting horizon (15–90 min). Results are reported as mean ± standard deviation over five runs with different random seeds. The best results are highlighted in bold, and the second-best results are underlined.

Models	MAE(°C)↓	RMSE(°C)↓	MAPE(%)↓	R^2 ↑	Inference Time (ms)↓
ANN	0.2339 ± 0.0039	0.3753 ± 0.0026	2.4780 ± 0.0801	0.8458 ± 0.0021	9.3286 ± 2.4000
LSTM	0.2218 ± 0.0021	0.3765 ± 0.0045	2.6112 ± 0.1819	0.8448 ± 0.0037	9.9328 ± 0.7668
BiGRU	0.1832 ± 0.0015	0.3256 ± 0.0025	2.5337 ± 0.1233	0.8839 ± 0.0018	14.3844 ± 2.5587
BiLSTM	0.2026 ± 0.0040	0.3501 ± 0.0030	2.5516 ± 0.0714	0.8658 ± 0.0023	13.3378 ± 3.4473
XGBoost	0.5366 ± 0.0000	1.2999 ± 0.0000	3.0885 ± 0.0000	0.7974 ± 0.0000	2.4126 ± 0.8619
ARIMA	0.2788 ± 0.0000	0.7156 ± 0.0000	1.3028 ± 0.0000	0.9386 ± 0.0000	852.96 ± 115.29
Attention-DNN	0.2095 ± 0.0013	0.3595 ± 0.0035	2.6338 ± 0.2282	0.8585 ± 0.0029	9.9850 ± 2.2738
Transformer	0.1833 ± 0.0014	0.3222 ± 0.0032	2.4655 ± 0.1110	0.8863 ± 0.0023	23.5661 ± 8.4440
CNN-BiLSTM-Attention	0.2521 ± 0.0074	0.4120 ± 0.0070	2.5043 ± 0.2153	0.8141 ± 0.0064	13.1164 ± 2.4145
BiTCN-Attention	0.1748 ± 0.0023	0.2945 ± 0.0019	2.2342 ± 0.1232	0.9050 ± 0.0012	12.2176 ± 3.1321
PatchTST	<u>0.0866 ± 0.0005</u>	<u>0.2242 ± 0.0006</u>	<u>1.1030 ± 0.0284</u>	<u>0.9450 ± 0.0003</u>	20.2512 ± 2.9296
FEDformer	0.1060 ± 0.0006	0.2372 ± 0.0005	1.2916 ± 0.0212	0.9384 ± 0.0003	32.9666 ± 10.6770
TFCG	**0.0574 ± 0.0024**	**0.1549 ± 0.0053**	**0.9844 ± 0.0346**	**0.9738 ± 0.0018**	18.7868 ± 5.4375



(a) Comparison with Attention-Based Models (b) Comparison with Non-Attention Models

Fig. 10. Box plots of residuals for the TFCG and other baseline models.

data center environments. Unlike existing forecasting methods that directly perform joint modeling across all variables, the proposed model combines channel-independent modeling with explicit time-frequency feature enhancement. Specifically, each rack is modeled individually based on its most relevant features, which helps reduce inter-channel interference while preserving intrinsic thermal dynamics. In addition, the STFT- and CNN-based time-frequency enhancement module enables the model to effectively capture both localized frequency patterns and temporal variations. Combined with the GRU-based temporal modeling framework for long-term dependency learning, the proposed method can simultaneously represent short-term fluctuations and long-term periodic behaviors. By jointly considering physical consistency, channel heterogeneity, and multi-scale temporal-frequency characteristics, the proposed model provides a more structured and effective solution for accurate and robust data center temperature prediction.

Prediction Stability Analysis. For clearer visualization, the residual box plots are divided into two subfigures in Fig. 10(a) comparison with attention-based models, and Fig. 10(b) comparison with non-attention models. To highlight the prediction stability of the models, Fig. 10 presents box plots of the residuals, defined as $y - \hat{y}$ (where y is the true value and $\hat{y}$ is the predicted value) for all models. It can be observed that the box height of the TFCG model is lower than those of the other models, and its median is closer to zero, indicating that TFCG exhibits the smallest prediction error variability and achieves the highest prediction stability.

Prediction Results Visualization. To provide a visual comparison of prediction performance, Fig. 11 presents the predicted results of the proposed TFCG model and several representative baseline models, including BiGRU, Transformer, BiTCN-Attention, PatchTST, and FEDformer. The blue curves denote the ground truth, while the red dashed curves represent the predicted values. It can be observed that TFCG consistently achieves closer alignment with the ground truth in terms of fitting accuracy, trend tracking, and responsiveness to extreme fluctuations. Compared with the other models, TFCG exhibits more precise prediction trajectories and better captures both global trends and local variations, demonstrating its superior overall prediction performance.

Convergence Analysis. Fig. 12 shows the training loss curves of all models over 200 batches under identical settings. For clarity, the models are divided into two groups: Fig. 12(a) presents the attention-based models, while Fig. 12(b) shows the non-attention models. As observed, TFCG rapidly converges to a low-loss region within the early stages and maintains a smooth and stable trajectory thereafter. In contrast, other models exhibit slower convergence and varying degrees of fluctuation. Although PatchTST shows relatively stable behavior initially, it converges to a higher loss with slight oscillations. Overall, TFCG achieves faster convergence, lower final loss, and better training stability than all baselines.

4.2.2. Model performance comparison under different settings

In this subsection, we compare the performance of the proposed model using a six-month subset of the dataset, various forecasting horizon settings, and another real-world air-cooled data center dataset.

Performance under Limited Data. In time-series forecasting tasks, the size of the training dataset plays an important role in model performance, as larger datasets typically provide richer temporal patterns and improve learning effectiveness. To further evaluate the performance of the proposed method under limited data availability, we conduct additional experiments using a six-month subset of the original dataset for comparison and analysis. The results on the six-month

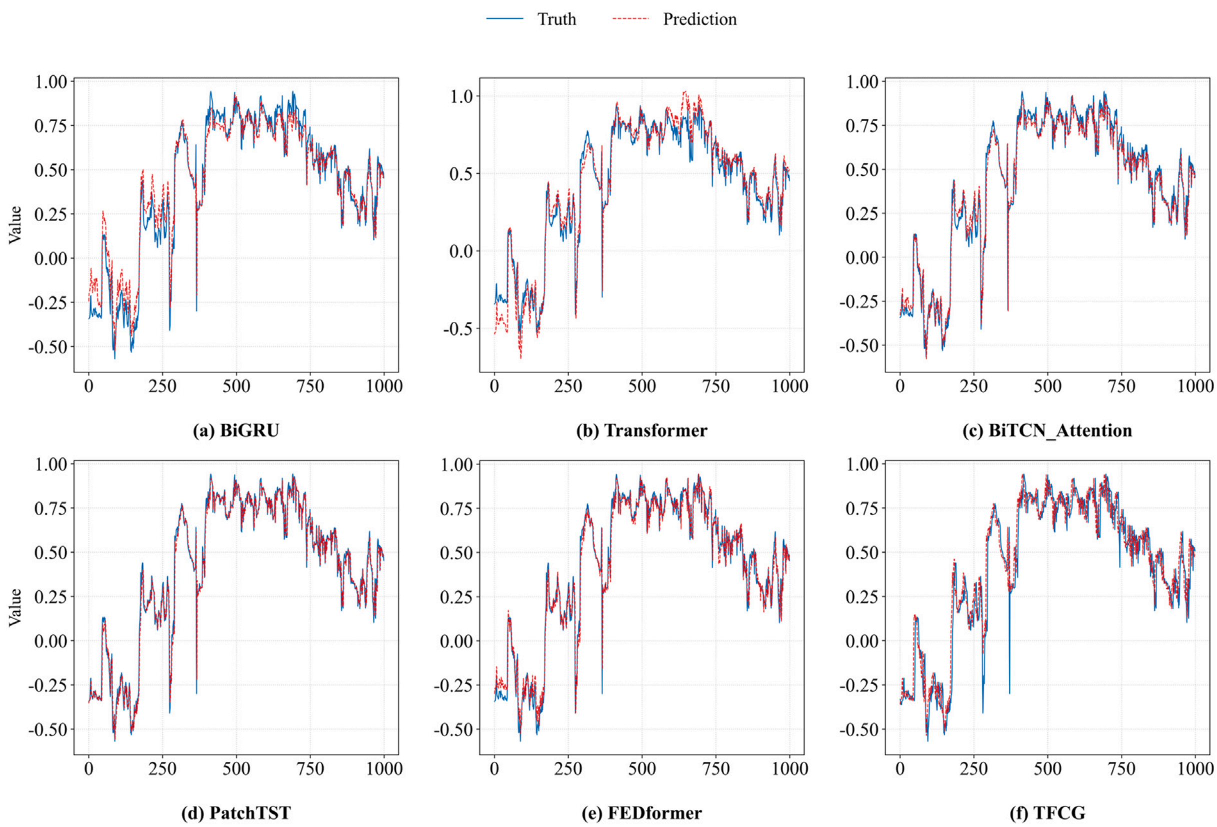
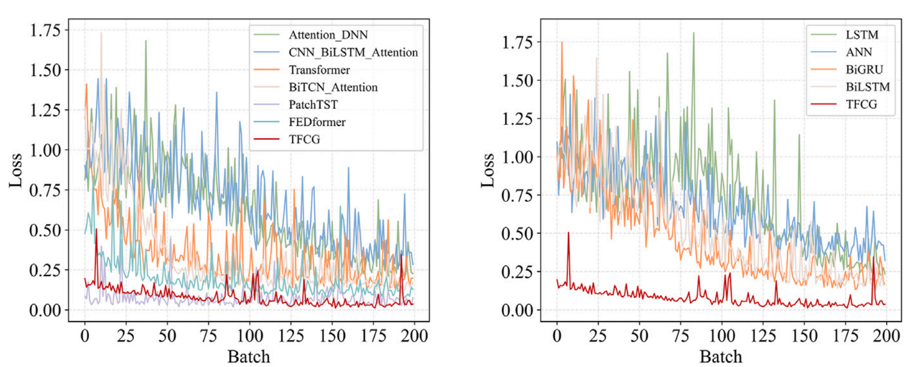


Fig. 11. Comparison of predicted and true values for all models on the same dataset.



(a) Comparison with Attention-Based models (b) Comparison with Non-Attention models

Fig. 12. Trends in loss values for the models during the training period.

subset of the dataset are presented in Table 2. It can be observed that the proposed TFCG model consistently outperforms all baseline methods, even when trained on a reduced data scale. Compared with strong baseline models such as PatchTST and FEDformer, the proposed TFCG model consistently achieves superior performance across all evaluation metrics. TFCG significantly reduces prediction errors while improving R^2, demonstrating its effectiveness under limited data conditions.

Performance under Different Forecasting Horizons. In time-series forecasting, the forecasting horizon is a critical factor, as longer horizons require the model to generalize over extended time spans while maintaining prediction accuracy. Therefore, experiments with different forecasting horizons are conducted to evaluate the proposed model. As shown in Fig. 13, the prediction accuracy of all models degrades as the forecasting horizon increases, in terms of MAE, RMSE, MAPE, and R^2. However, the proposed TFCG model consistently achieves the best performance and exhibits a much more gradual degradation trend. This indicates that TFCG is more robust to increasing prediction difficulty and is better able to maintain stable performance over long-term forecasting horizons.

These results demonstrate that the proposed model effectively integrates time-domain and frequency-domain feature extraction, enabling it to capture long-term temporal dependencies. In addition, the adoption of a channel-independent strategy allows the model to learn the most relevant features for each rack individually. As a result, the model can better adapt to the intrinsic characteristics of data center thermal dynamics and achieve more accurate and stable predictions under varying forecasting horizons.

Table 2
Prediction performance comparison of different models on a 6-month subset of the dataset. The best values for each metric are highlighted in bold, and the second-best values are underlined.

Models	MAE(°C)↓	RMSE(°C)↓	MAPE(%)↓	R^2 ↑	Inference Time (ms)↓
ANN	0.430803	0.618182	3.86444	0.628089	36.4954
LSTM	0.384151	0.579255	4.76030	0.673452	40.6884
BiGRU	0.313824	0.517357	3.83358	0.739512	39.1864
BiLSTM	0.347316	0.543478	4.25272	0.712544	43.0694
ARIMA	0.234512	0.544477	**1.06790**	0.874306	1775.18
XGBoost	0.422990	0.807016	1.91119	0.723865	1.6060
Attention-DNN	0.397823	0.592650	5.14153	0.658175	35.4697
CNN-BiLSTM-Attention	0.402952	0.603452	4.42206	0.645601	40.8520
Transformer	0.323341	0.519548	5.22139	0.737301	66.1523
BiTCN-Attention	0.327864	0.523403	3.89308	0.733388	40.8681
PatchTST	0.137724	0.289769	3.00528	0.918283	45.3974
FEDformer	0.176958	0.355365	3.57699	0.877099	78.9435
TFCG	**0.092604**	**0.213788**	1.19979	**0.955582**	49.0887

Table 3
Performance comparison on another publicly available air-cooled data center dataset. The best results are highlighted in bold, and the second-best results are underlined.

Models	MAE(°C)↓	RMSE(°C)↓	MAPE(%)↓	R^2 ↑	Inference Time (ms)↓
ANN	0.237834	0.364785	4.415923	0.859897	4.4795
LSTM	0.201530	0.322404	3.723218	0.890561	4.7934
BiGRU	0.137400	0.228596	2.932540	0.944981	8.2817
BiLSTM	0.149244	0.249432	3.218465	0.934495	6.8020
XGBoost	0.095103	0.344902	**0.429038**	**0.978207**	1.2883
ARIMA	0.120555	0.463328	0.544512	0.960673	2789.06
Attention-DNN	0.183504	0.291717	3.697642	0.910403	3.2201
Transformer	0.121569	0.207871	2.686519	0.954506	12.7272
CNN-BiLSTM-Attention	0.170803	0.270290	4.299303	0.923081	5.6471
BiTCN-Attention	0.124852	0.211587	2.687767	0.952865	8.7218
PatchTST	0.132545	0.246906	2.369303	0.935815	18.2774
FEDformer	0.130799	0.220152	1.963857	0.948971	21.3754
TFCG	**0.072660**	**0.159675**	0.802104	0.973158	10.3193

Performance under Different Datasets. To further validate the generalization capability of the proposed model, we additionally evaluate it on another publicly available air-cooled data center dataset [34]. In this experiment, temperature and airflow velocity variables are mainly utilized, and all variables are temporally aligned with a sampling interval of 10 s. As shown in Table 3, the proposed TFCG model consistently achieves the best overall prediction performance across most evaluation metrics. Specifically, TFCG obtains the lowest MAE and RMSE values, indicating superior accuracy and stability in temperature forecasting under different cooling environments and data distributions. Compared with recent deep learning models such as PatchTST and FEDformer, the proposed model still demonstrates clear advantages, which further verifies the effectiveness of the channel-independent modeling strategy and the time–frequency enhancement mechanism. These results demonstrate that the proposed method exhibits strong robustness and generalization capability across different data center thermal management scenarios.

4.2.3. Ablation studies

To quantify the contributions of the core components to overall model performance, an ablation study was conducted by systematically removing individual modules, including the STFT-based frequency enhancement, channel-independent mechanism, feature selection module,

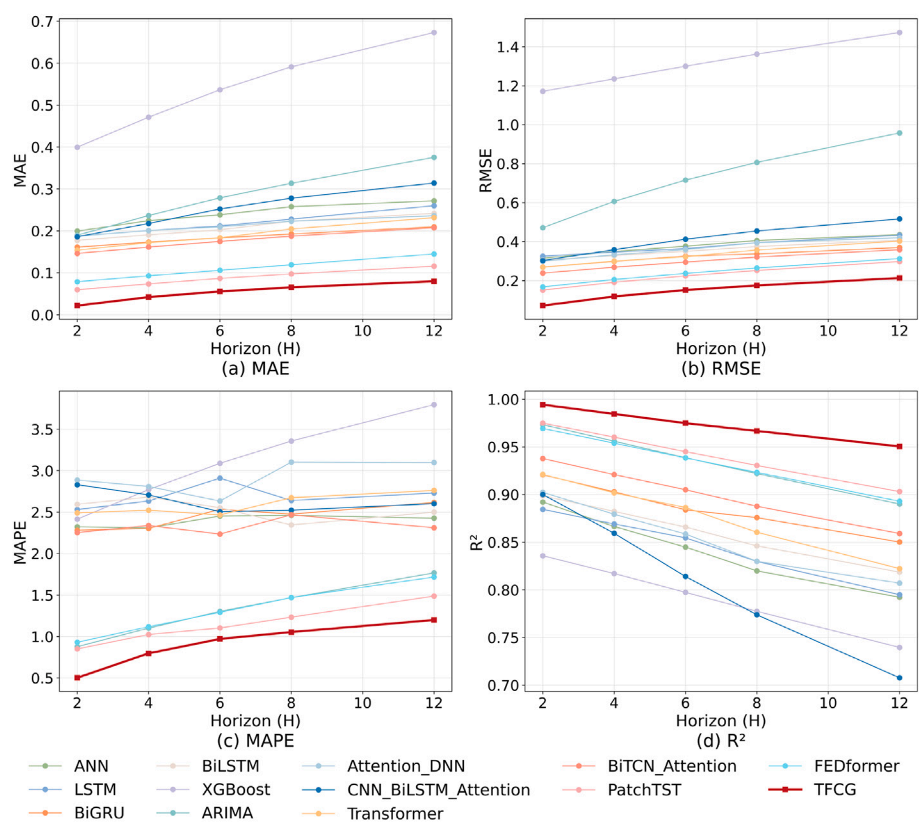


Fig. 13. Comparison of model evaluation results across different prediction horizons.

Table 4
Results of the ablation study. The best values for each metric are highlighted in bold, and the second-best values are underlined.

Models	MAE(°C)↓	RMSE(°C)↓	MAPE(%)↓	R^2 ↑	Inference Time (ms)↓
TFCG	**0.055105**	**0.149993**	<u>0.9460</u>	**0.975438**	29.2672
TFCG-TF	0.064518	0.188318	1.0495	0.961283	21.6343
TFCG-CI	0.056491	0.155731	0.9776	<u>0.973523</u>	17.9869
TFCG-FS	0.057499	0.154904	**0.6852**	0.969326	32.8856
TFCG-C	0.062270	0.168181	1.0474	0.969121	15.2252
TFC-LSTM	0.061987	0.168765	1.0431	0.968905	33.4293
TFC-TCN	0.064518	0.188318	1.0495	0.961283	24.3115
Wavelet-CG	<u>0.055407</u>	<u>0.150991</u>	0.9609	0.951104	120.8066

and CNN-based temporal enhancement. In addition, to further validate the effectiveness of the GRU-based temporal modeling, we replaced it with alternative sequence models, namely LSTM and TCN. We also conducted an additional experiment by replacing the STFT-based frequency enhancement module with wavelet analysis to further evaluate the effectiveness and computational efficiency of different frequency-domain analysis methods.

The methods used in the ablation study are summarized as follows:

- TFCG: The full proposed model.
- TFCG-TF: The frequency-domain enhancement module based on STFT is removed.
- TFCG-CI: The channel-independent module is removed.
- TFCG-FS: The feature selection module is removed.
- TFCG-C: The time-domain enhancement module based on CNN is removed.
- TFC-LSTM: The GRU is replaced with LSTM.
- TFC-TCN: The GRU is replaced with TCN.
- Wavelet-CG: The STFT is replaced with Wavelet.

As shown in Table 4, removing or modifying each component leads to distinct performance variations. Notably, the variant without the feature selection module (TFCG-FS) achieves a lower MAPE compared with the full model, indicating that excluding feature selection can sometimes improve performance in absolute-percentage-based metrics. This suggests that the feature selection mechanism may introduce beneficial regularization in certain cases, while also potentially suppressing subtle features that contribute to error-sensitive metrics. When replacing GRU with LSTM, the model exhibits noticeable performance degradation across all evaluation metrics, indicating that the GRU-based temporal modeling framework is more effective for capturing long-term thermal dependencies in this task. In contrast, the TCN-based variant shows more noticeable performance degradation, further highlighting the effectiveness of GRU in capturing long-term temporal dependencies in this task. Other ablation variants, obtained by removing individual modules, also perform worse than the full model, confirming the importance of each component.

In addition, replacing STFT with wavelet analysis does not lead to significant degradation in evaluation metrics such as MAE and RMSE. However, the inference time of wavelet analysis is approximately four times that of STFT, indicating that STFT provides a better balance between prediction performance and computational efficiency for real-time data center thermal management. Other ablation variants, obtained by removing individual modules, also perform worse than the full model, confirming the importance of each component. Overall, the ablation study demonstrates that each module contributes uniquely to the final performance. The integration of time-frequency feature enhancement, GRU-based long-term dependency modeling, channel-independent processing, and feature selection enables the proposed TFCG model to achieve high-precision predictions.

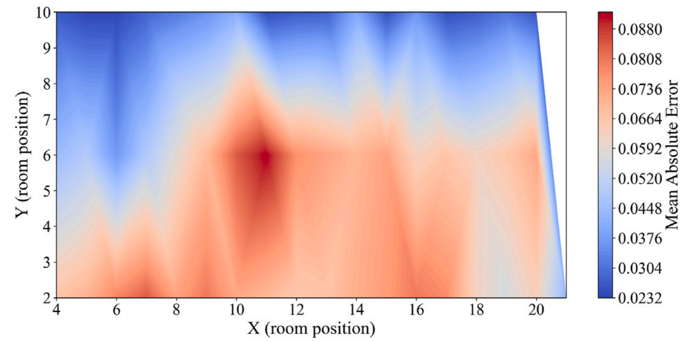


Fig. 14. Heatmap of the spatial distribution of prediction errors.

4.2.4. Spatial distribution characteristics of prediction errors

To further investigate the spatial distribution of prediction errors, a heatmap of absolute errors for each rack was generated, as shown in Fig. 14, overlaid with the data center layout from Fig. 9. The results reveal a clear spatial dependency of prediction errors across the data center. In general, racks located in the central area of the room exhibit relatively higher error levels, whereas racks near the room boundaries, particularly those close to air conditioning units, show significantly lower prediction errors. This pattern suggests that temperature dynamics are more stable in regions directly influenced by cooling sources, enabling the model to capture their behavior more accurately. In contrast, central racks are more strongly affected by cold air attenuation and complex hot-cold airflow mixing, resulting in more heterogeneous thermal conditions and slightly lower prediction accuracy.

5. Conclusion

This study proposes the TFCG model to address the challenges of multi-scale dynamic modeling and cross-channel noise interference in data center thermal prediction. The proposed method introduces a time-frequency enhancement module to jointly extract temporal and spectral features from multi-dimensional monitoring signals, enabling the characterization of complex temperature variation patterns. For temporal dependency modeling, a Gated Recurrent Unit (GRU) is employed to capture long-range dependencies, facilitating accurate rack temperature forecasting. In addition, a channel-independent modeling strategy is adopted to model each server rack separately, while multi-channel feature representations are incorporated to improve both prediction accuracy and robustness. Experimental results on real-world data center rack temperature datasets demonstrate that the proposed method significantly outperforms existing mainstream approaches in prediction accuracy.

In summary, the proposed TFCG model provides an effective solution for data center temperature prediction, contributing to the efficient energy management of data center cooling systems.

CRediT authorship contribution statement

Yu Li: Writing – original draft. **Weiwei Lin:** Writing – review & editing, Funding acquisition. **Jianpeng Lin:** Writing – review & editing. **Zhiyu Yan:** Writing – review & editing. **Junxian Shen:** Writing – review & editing. **Xiumin Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.asoc.2026.115789.

Data availability

Data will be made available on request.

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